

Paper 2 - Set 02 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	E	Indices and surds
2	C	Trigonometric functions and equations
3	A	Exponentials and logarithms
4	H	Sequences and series
5	A	Triangles, radians and sectors
6	A	Sequences and series
7	D	Chains of reasoning
8	B	Chains of reasoning
9	F	Chains of reasoning
10	C	The logic of arguments
11	C	Mathematical proof
12	F	Differentiation
13	A	Mathematical proof
14	E	The logic of arguments
15	E	Identifying errors in proofs
16	F	Quadratic functions
17	E	Triangles, radians and sectors
18	B	Graphs of functions
19	B	The logic of arguments
20	E	Mathematical proof

Worked solutions

Question 1 IND-13 · Indices and surds · spec MM1.1

Simplify $\left(\frac{a^{\frac{1}{2}}b^{-2}}{a^{-1}b^{\frac{1}{2}}}\right)^4$.

Answer: E

Inside the bracket: $a^{\frac{1}{2}+1}b^{-2-\frac{1}{2}} = a^{\frac{3}{2}}b^{-\frac{5}{2}}$. Raise to the 4th power: a^6b^{-10} .

Question 2 TRG-18 · Trigonometric functions and equations · spec MM4.3

Find the value of $\tan 30^\circ \tan 60^\circ + \sin^2 45^\circ$.

Answer: C

$\tan 30^\circ \tan 60^\circ = \frac{1}{\sqrt{3}} \cdot \sqrt{3} = 1$, $\sin^2 45^\circ = \frac{1}{2}$. Total $\frac{3}{2}$.

Question 3 LOG-10 · Exponentials and logarithms · spec MM5.3

How many real solutions does the equation $4^x + 2^{x+1} - 8 = 0$ have?

Answer: A

$y = 2^x$: $y^2 + 2y - 8 = (y+4)(y-2) = 0$. $y = -4$ is impossible, $y = 2$ gives $x = 1$. One solution.

Question 4 X2-23 · Sequences and series · spec MM2.2

An arithmetic sequence has first term a and common difference d , both non-zero. Which of the following statements is/are true?

- I If $d > 0$ then every term after some point is positive.
- II The sum of the first n terms is a quadratic function of n .
- III If a and d are integers then S_n is an integer for every n .

Answer: H

I: terms $a + (n-1)d$ eventually exceed any bound. II: $S_n = \frac{n}{2}(2a + (n-1)d)$ is quadratic in n . III: S_n is a sum of integers.

Question 5 X2-15 · Triangles, radians and sectors · spec MM4.1

In triangle ABC , P : ' $a^2 = b^2 + c^2$ '. Q : 'angle $A = 90^\circ$ '. Which one of the following is true?

Answer: A

Pythagoras and its converse: the cosine rule gives $\cos A = \frac{b^2 + c^2 - a^2}{2bc}$, which is 0 exactly when $a^2 = b^2 + c^2$.

Question 6 X2-04 · Sequences and series · spec MM2.3

A geometric series has first term $a \neq 0$ and common ratio r . P : ' $-1 < r < 1$ '. Q : 'the series has a sum to infinity'. Which one of the following is true?

Answer: A

A geometric series converges if and only if $|r| < 1$, which is exactly $-1 < r < 1$.

Question 7 RSN-22 · Chains of reasoning · spec Prf5

A 4×4 grid of squares is coloured so that every row and every column contains exactly two black squares. How many black squares are there in the grid?

Answer: D

Each of the 4 rows contains exactly 2 black squares: 8 in total.

Question 8 RSN-07 · Chains of reasoning · spec Prf5

The digits 1 to 9 are placed in a 3×3 grid so that every row and every column has the same sum. What is that sum?

Answer: B

The three rows together contain every digit once, so $3 \times (\text{row sum}) = 1 + 2 + \dots + 9 = 45$, giving 15.

Question 9 RSN-16 · Chains of reasoning · spec Prf5

The sum of 5 consecutive odd numbers is S . Which of the following is always a factor of S ?

Answer: F

$(m-4) + (m-2) + m + (m+2) + (m+4) = 5m$ where m is the middle (odd) number; $5m$ is odd, so 10 is never a factor.

Question 10 LGC-04 · The logic of arguments · spec Arg1

Consider the statement about a real number x : **if $x^2 = x$, then $x = 1$.** Which one of the following is true?

Answer: C

$x = 0$ satisfies $x^2 = x$ but not $x = 1$, so the statement is false. The converse, 'if $x = 1$ then $x^2 = x$ ', is true.

Question 11 PRF-19 · Mathematical proof · spec Prf1

A student proves that $\sqrt{4}$ is irrational by copying the standard proof for $\sqrt{2}$: 'Suppose $\sqrt{4} = p/q$ in lowest terms; then $p^2 = 4q^2$, so p^2 is even, so p is even, $p = 2k$; then $4k^2 = 4q^2$, so $k^2 = q^2 \dots$ ' At which point does the argument break down compared with the $\sqrt{2}$ proof?

Answer: C

For $\sqrt{2}$ the step $p^2 = 2q^2$ leads to $2k^2 = q^2$, making q even. Here $k^2 = q^2$ carries no parity information, so the contradiction never appears - as expected since $\sqrt{4} = 2$.

Question 12 X2-16 · Differentiation · spec MM6.3

The function f has $f'(x) = (x-1)(x-3)^2$. Which of the following statements is/are true?

- I f has a local minimum at $x = 1$.
- II f has a local maximum at $x = 3$.
- III f is increasing for all $x > 1$.

Answer: F

f' changes from negative to positive at $x = 1$ (minimum). At $x = 3$ the squared factor keeps $f' \geq 0$ on both sides, so no turning point: II is false, and $f' \geq 0$ for $x > 1$ (zero only at 3) makes III true.

Question 13 PRF-12 · Mathematical proof · spec Prf2

For a positive integer n it is known that: (i) if n is a multiple of 4 then n is not a multiple of 6; (ii) n is a multiple of 12 or n is a multiple of 4. What can be deduced?

Answer: A

If n were a multiple of 12 it would be a multiple of both 4 and 6, contradicting (i). So by (ii) n is a multiple of 4, and by (i) not a multiple of 6.

Question 14 LGC-20 · The logic of arguments · spec Arg1

Which of the following statements is/are logically equivalent to 'if it is raining, then the ground is wet'?

- I It is not raining, or the ground is wet.
- II If the ground is not wet, then it is not raining.
- III If the ground is wet, then it is raining.

Answer: E

' $A \Rightarrow B$ ' is equivalent to 'not A , or B ' (I) and to its contrapositive (II). III is the converse, which is not equivalent.

Question 15 ERR-17 · Identifying errors in proofs · spec Err1

Claim: if $f'(a) = 0$ and $f''(a) = 0$ then f has a point of inflexion at $x = a$.

Which of the following shows the claim is false?

Answer: E

x^4 has $f'(0) = f''(0) = 0$ but a minimum at 0, not an inflexion. (x^3 does have an inflexion; x^2 has $f''(0) \neq 0$.)

Question 16 QUA-17 · Quadratic functions · spec MM1.3

The quadratic equation $ax^2 + bx + c = 0$ (with $a \neq 0$) has real coefficients. Which of the following statements is/are always true?

- I If $ac < 0$, the equation has two distinct real roots.
- II If $b = 0$, the equation has no real roots.
- III If the equation has a repeated root, then $b^2 = 4ac$.

Answer: F

I: $b^2 - 4ac > 0$ whenever $ac < 0$, so true. II: $x^2 - 4 = 0$ has $b = 0$ and real roots, so false. III: a repeated root means the discriminant is zero, so true.

Question 17 TRI-12 · Triangles, radians and sectors · spec MM4.1

A triangle has sides a, b, c with $a^2 + b^2 < c^2$. Which of the following statements is/are true?

- I The angle opposite c is obtuse.
- II c is the longest side.
- III The triangle cannot be isosceles.

Answer: E

Cosine rule: $\cos C = \frac{a^2 + b^2 - c^2}{2ab} < 0$, so C is obtuse (I). The largest angle is opposite the longest side, so c is longest (II). III is false: $a = b = 1, c = 1.5$ works.

Question 18 X2-09 · Graphs of functions · spec MM8.7

The graphs of $y = f(x)$ and $y = g(x)$ meet at exactly three points. Which of the following statements must be true?

- I The equation $f(x) = g(x)$ has exactly three real solutions.
- II The graph of $y = f(x) - g(x)$ crosses the x -axis exactly three times.
- III $f(x) - g(x)$ is a polynomial of degree at least 3.

Answer: B

I is the definition of an intersection. II fails because the difference could touch the axis rather than cross it at one of the points. III fails because f and g need not be polynomials at all (e.g. $\sin x$ and a line).

Question 19 LGC-05 · The logic of arguments · spec Arg3

Which of the following statements is/are true?

- I For every real number x , there exists a real number y such that $y > x$.
- II There exists a real number y such that for every real number x , $y > x$.
- III For every real number x , there exists a real number y such that $xy = 1$.

Answer: B

I is true (take $y = x + 1$). II claims a single largest real number exists: false. III fails for $x = 0$.

Question 20 PRF-18 · Mathematical proof · spec Prf2

Every element of a set S of integers is either a multiple of 3 or a multiple of 5. Every multiple of 3 in S is negative. S contains the number 10. Which of the following statements must be true?

- I S contains a positive multiple of 5.
- II S contains no positive multiple of 3.
- III S contains a negative number.

Answer: E

$10 \in S$ is a positive multiple of 5 (I). Multiples of 3 in S are negative, so none is positive (II). III fails: $S = \{10\}$ satisfies everything.